FAIR-CSAR: A Benchmark Dataset for Fine-Grained Object Detection and Recognition Based on Single-Look Complex SAR Images

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***Abstract*— Object detection and recognition (OD&R) based on deep learning is a hot topic in the application of synthetic aperture radar (SAR). These methodologies based on deep learning are inherently data-driven, which means that their performance is subjected to the corresponding datasets. Although existing datasets have included some common targets collected from real-valued intensity SAR images, there still exist some limitations in terms of quantity, categories, diversities, and data domain. Hence, it is urgent to establish a large-quantity benchmark for fine-grained OD&R on complex-valued SAR images, which contains rich signal-domain features well coupled with classical physical modeling. In addition, considering the unique imaging characteristics and diverse imaging conditions, some important attribute information, such as incidence and attitude angles, is necessary to be attached. In this article, we propose a novel benchmark dataset with more than 340k instances for fine-grained OD&R based on single-look complex (SLC) SAR images, which is named FAIR-CSAR. We collected complex-valued SAR images with a resolution of 1–5 m from 175 entire images of Gaofen-3 covering 32 cities and multiple sea areas worldwide. All instances in the FAIR-CSAR are annotated by oriented bounding boxes (OBBs), covering five major categories and 22 subcategories. Compared with existing datasets dedicated to OD&R, the FAIR-CSAR dataset has four particular advantages: 1) it contains complex-valued SAR images from various acquisition modes and polarization modes, including full-scale signal-domain features for object recognition; 2) it is much larger than other existing OD&R datasets in terms of quantity of instances; 3) it provides more fine-grained category annotation and more detailed attribute information; and 4) it provides more challenging images with some common imaging phenomena, such as speckle noise and azimuth ambiguities. To establish a baseline adapted for SLC SAR images, a multidomain feature extraction and fusion network (MDNet) is proposed as a novel framework to mine detailed information underlying various domains. A series of state-of-the-art (SOTA) algorithms are applied on the FAIR-CSAR to build the fine-grained OD&R benchmark. Experimental results indicate that FAIR-CSAR is closer to practical application and more challenging than existing datasets for SAR images.**

***Index Terms*— Benchmark dataset, object detection and recognition (OD&R), single-look complex (SLC), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is a common remote sensing payload that actively emits electromagnetic waves and receives its echo to acquire Earth surface scattering information. Contributed to its waveband being less affected by atmospheric attenuation, SAR can achieve global monitoring and information collection all day and all weather [1], [2]. The energy of fully collected information scattered by ground targets disperses within the complex-valued echo, lacking the remarkable features to distinguish various targets [3]. In order to achieve clear features with sufficient resolution and signal-to-noise ratio (SNR), the echo energy needs to undergo specific processes, such as pulse compression [4] and range cell migration correction (RCMC) [5] in the complex-valued signal domain. The processed results are usually presented as real-valued intensity images, which abandon some physical signal-domain features but retain the most intuitive, basic visual features for image understanding and interpretation [6].

In light of the unique merits of SAR systems, object detection and recognition (OD&R) for SAR images has become a crucial component of image understanding and interpretation, widely applied for commercial and military fields [7], [8]. Currently, almost all application-oriented OD&R algorithms are based on deep learning, which appears robust generalization and accurate capabilities in computer vision [9],

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[10], [11]. Deep learning has also made remarkable strides in OD&R on SAR images [12], [13], [14]. It can effectively extract crucial features, address challenges such as small target detection and intraclass imbalance, and enhance detection accuracy and efficiency. These methodologies based on deep learning are inherently data-driven [15]; hence, the characteristics of datasets directly constrain the boundaries of algorithm applications. For example, the generalization ability and the recognition scope are subjected to the quantity and the granularity of the datasets, respectively. For the same reason, the richness of feature collection and utilization also depends on the choice of information domain of datasets. It means that a comprehensive benchmark dataset is crucial for the development of algorithm applications.

Recently, some excellent benchmark datasets have significantly accelerated the development of OD&R algorithms applied in various remote sensing images. For optical remote sensing images, the dataset for object detection in aerial images (DOTA) [16] and FAIR1M [17] datasets have significantly advanced research of OD&R applied in complex scenarios for its large-quantity fine-grained annotations. The instances of them are annotated in the form of oriented bounding boxes (OBBs), which promote the development of oriented object detection algorithms [18]. As for SAR remote sensing images, some classical datasets are proposed. One of the famous and high-quality datasets is the SAR ship detection dataset (SSDD) dataset [19], which supports lots of early studies on ship detection. In addition, datasets such as SAR-Aircraft-1.0 [20] and SAR Airport Dataset (SAD) [21] also make contributions to the studies for other categories, such as aircraft and airports. However, nearly all of these datasets are based on SAR intensity images, losing valuable signal-domain information coupled with classical physical modeling. This constrains the application domain of the developed algorithms and also limits their further performance improvement, especially for fine-grained OD&R algorithms. Furthermore, compared with the well-developed optical benchmark datasets, the existing SAR datasets also reveal some common shortcomings that urgently need to be improved.

A. Dataset Construction Needs to Further Focus on Complex-Valued Images

After imaging, nearly all scattering features are hidden in the single-look complex (SLC) image, while the most intuitive visual features, such as contour and texture, can be easily acquired in its subset, the real-valued intensity image, as shown in Fig. 1. Contributed to the representation of these visual features similar to that in optical images, the object detection algorithms suitable for optical images can be easily transferred for the relevant application on the SAR intensity images. As a result, most of the existing SAR algorithms and datasets are constructed on real-valued intensity images [19], [22], [23]. However, due to coherent imaging mechanisms and other factors, SAR targets appear unusual discrete, multicentric visual features [24], [25], which are further blurred by speckle noise [26], [27]. These factors further limit the performance of the existing algorithms on the fine-grained OD&R application. Fortunately, the scattering features are fully carried in SAR complex-valued images, where other signal-domain features can also be extracted and further used for performance improvement, including range-Doppler features, 2-D frequency features, time–frequency (TF) features, and so on. These extra features from complex-valued images assist in constructing a hybrid feature space with a larger margin among various categories, as shown in Fig. 1. Therefore, it is necessary to construct a fine-grained OD&R dataset based on complex-valued images, which further promotes the development of the relevant application.

B. Annotation Granularity Needs To Be Refined

Due to differences caused by the unique scattering mechanism, the visual features of SAR objects appear discrete and fragmented, which results in the significant difficulty of SAR object identification [32]. Therefore, up until now, there is still a lack of datasets with hierarchical fine-grained annotation. The existing datasets, of which annotation information is either for a single category or for coarse-grained multiclass categories (as shown in Fig. 2), are not enough to support SAR fine-grained object recognition. Moreover, SAR object features vary with the geometric relationship between satellite and targets [33], some imaging parameters, and so on. To explore the deep association between features and parameters, it is urgent to annotate other necessary attached information, such as target attitude angle, incidence angle, and center frequency. Therefore, the annotation granularity of category information and other attached information needs to be further enriched.

C. Quantity of Instances Can Be Further Expanded

Different from optical features, the intuitive visual features of object instances in SAR are sensitive to imaging conditions, such as incidence angle, center frequency, and resolution. These result in larger intraclass differences but smaller interclass differences [31]. Therefore, the SAR feature space is larger, which needs to be characterized by a sufficient quantity of SAR instances. In addition, in order to verify the generalization ability and avoid overfitting, the size of the SAR dataset should be kept at a higher level. However, almost all existing SAR datasets have less than 50k image slices with insufficient instances. The quantity of instances in the SAR dataset needs to be expanded urgently.

D. Dataset Needs To Be More Diverse and Challenging

In practice, there are some unique obstacles in SAR OD&R. For example, azimuth ambiguities and speckle noise can lead to false alarms and missed detection [34], [35]. Object defocusing caused by motion can make recognition more difficult [36]. However, most existing datasets only consist of high-quality clear images acquired in common imaging conditions, failing to verify the relevant application close to the actual scenes. Hence, it is necessary to supplement some multiscene hard examples containing the abovementioned factors to keep the dataset diverse and challenging.

To address the above issues, we propose a benchmark dataset, named FAIR-CSAR, for fine-grained OD&R based

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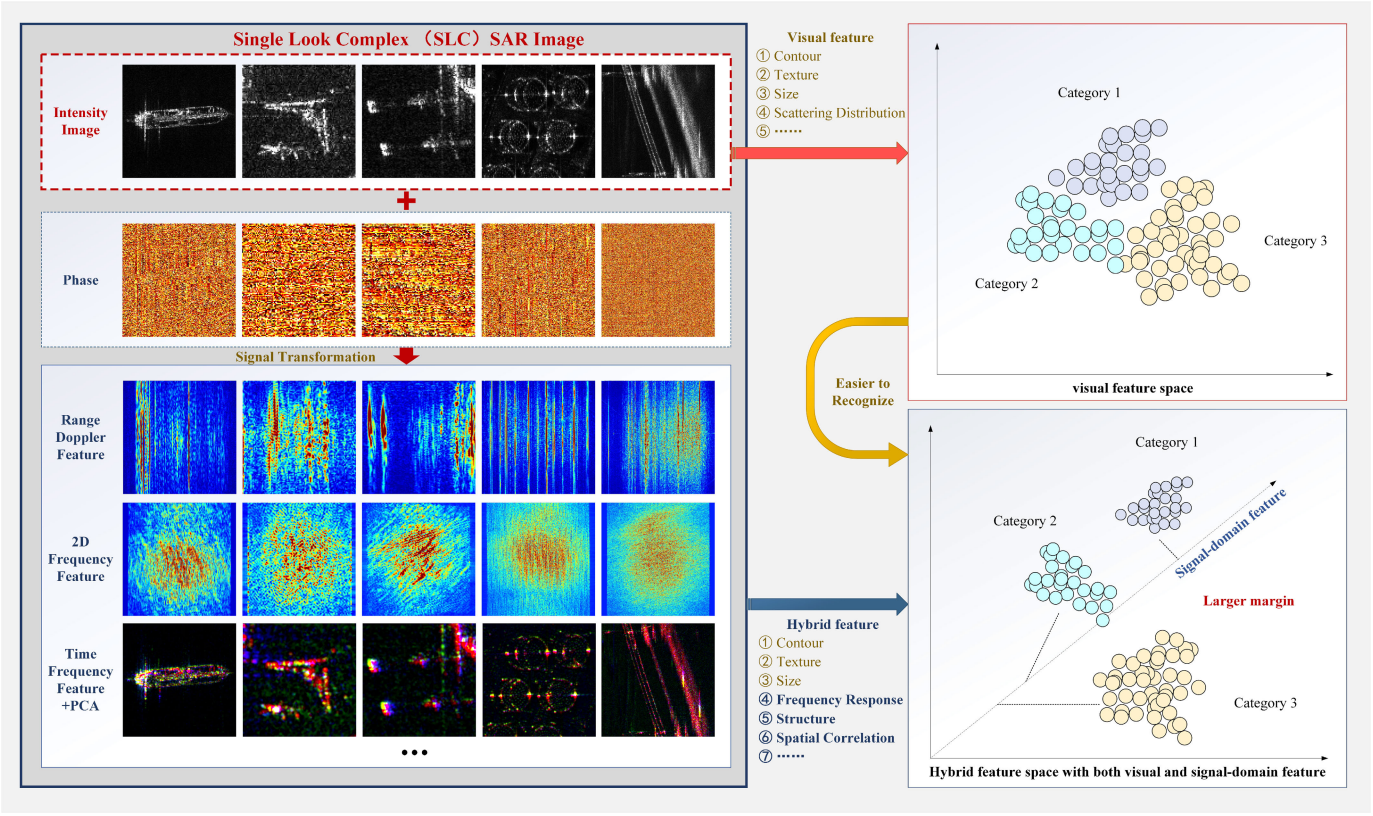


Fig. 1. Illustration of the advantage of the SLC SAR image on the fine-grained OD&R tasks. Different from intensity images only offering common visual features, SLC SAR images can acquire more signal-domain features from various transform domains, including range-Doppler features, 2-D frequency features, TF features, and so on. With the supplement of signal-domain features, the targets, which are difficult to distinguish in the pure visual feature space, have the potential to be classified in the hybrid feature space.

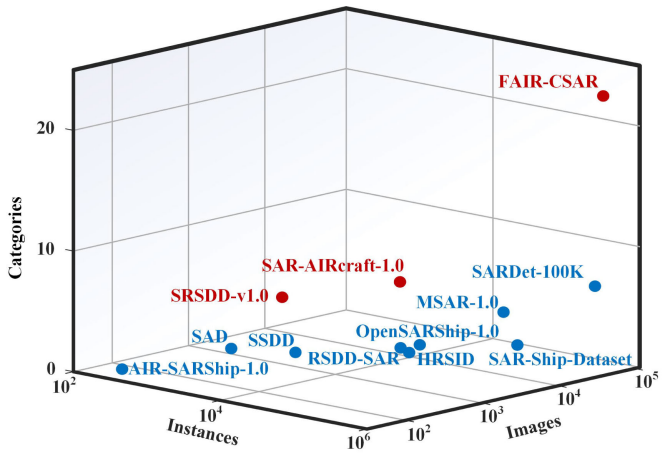


Fig. 2. Multidimensional comparison for typical SAR object detection datasets in remote sensing. Blue and red points stand for the generic datasets and the fine-grained datasets, respectively.

on SLC SAR images from Gaofen-3. Some representative examples of images and their annotations are shown in Fig. 3. The FAIR-CSAR dataset with fine-grained annotation totally contains over 340k instances, which are acquired from 175 entire images of Gaofen-3 covering 32 cities located in various regions (see Fig. 4), such as East Asia, Southeast Asia, South Asia, Europe, Middle East, and America. All instances are annotated with OBBs, covering five major categories and 22 subcategories in total. To the best of our knowledge, the FAIR-CSAR dataset is the largest fine-grained OBB dataset with SAR detailed attached information based on complex-valued SAR images.

With fine-grained annotation and massive volume, FAIR-CSAR aims to provide a comprehensive experimental foundation for the verification of SAR OD&R algorithms. Furthermore, it provides not only SLC images with richer information and more possibilities but also abundant attached information including system parameters, processing parameters, and so on. These unique advantages make FAIR-CSAR capable of more in-depth research combined with imaging processing.

In order to standardize the development of fine-grained OD&R on the FAIR-CSAR dataset, we build a benchmark using representative algorithms. Also, we propose a multidomain feature extraction and fusion network (MDNet) as a novel framework to mine detailed information underlying SLC SAR images, which can serve as a reference to provide a starting point for further exploration.

In summary, the proposed FAIR-CSAR benchmark dataset aims to provide a complex-valued dataset with large-quantity detailed annotations for research on fine-grained SAR detection and recognition. We hope that with the help of FAIR-CSAR, an increasing number of novel physics-aware, trustworthy algorithms coupled with SAR characteristics can be proposed and applied in the SAR image understanding

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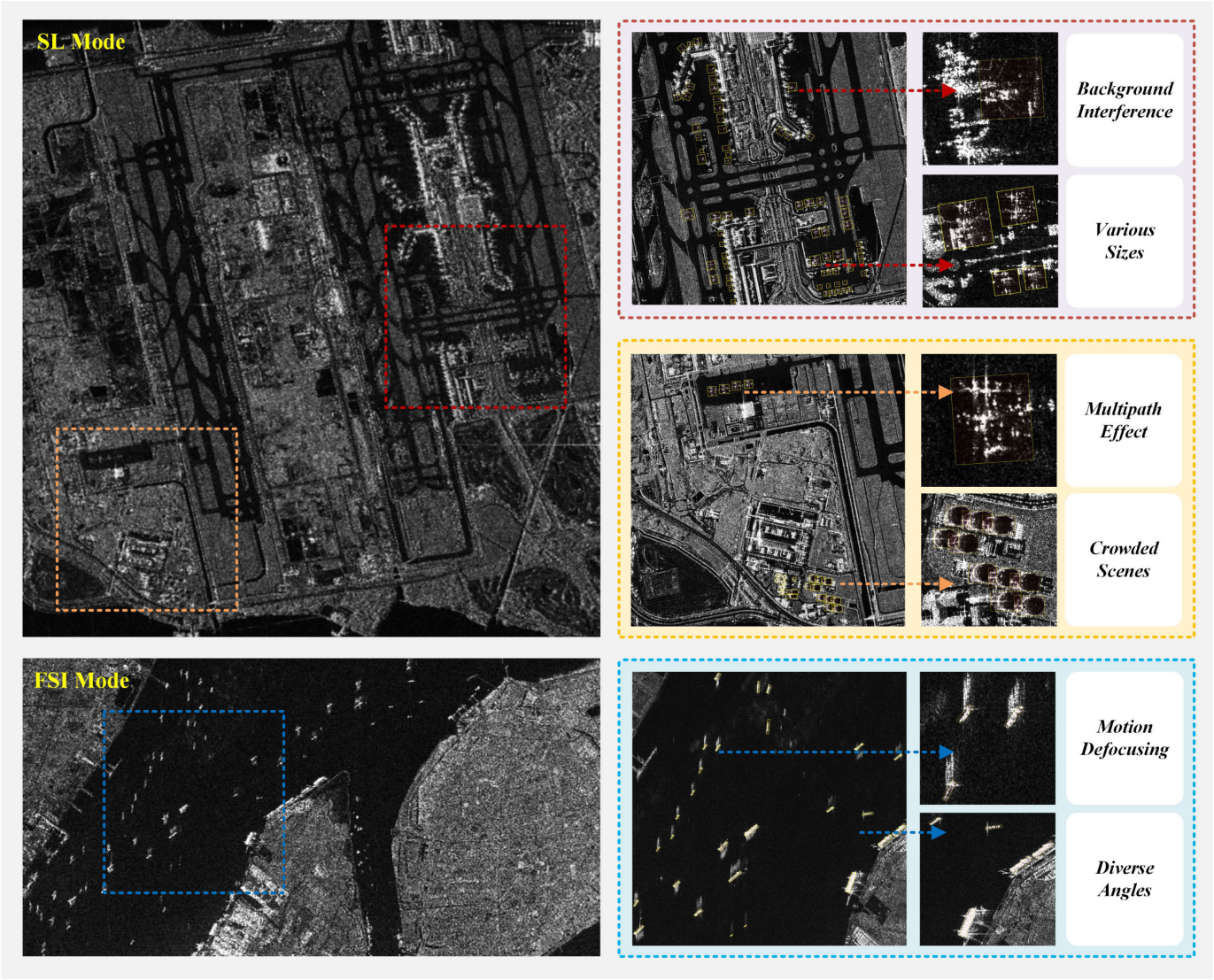


Fig. 3. Representative examples of images and corresponding annotations in the FAIR-CSAR dataset.

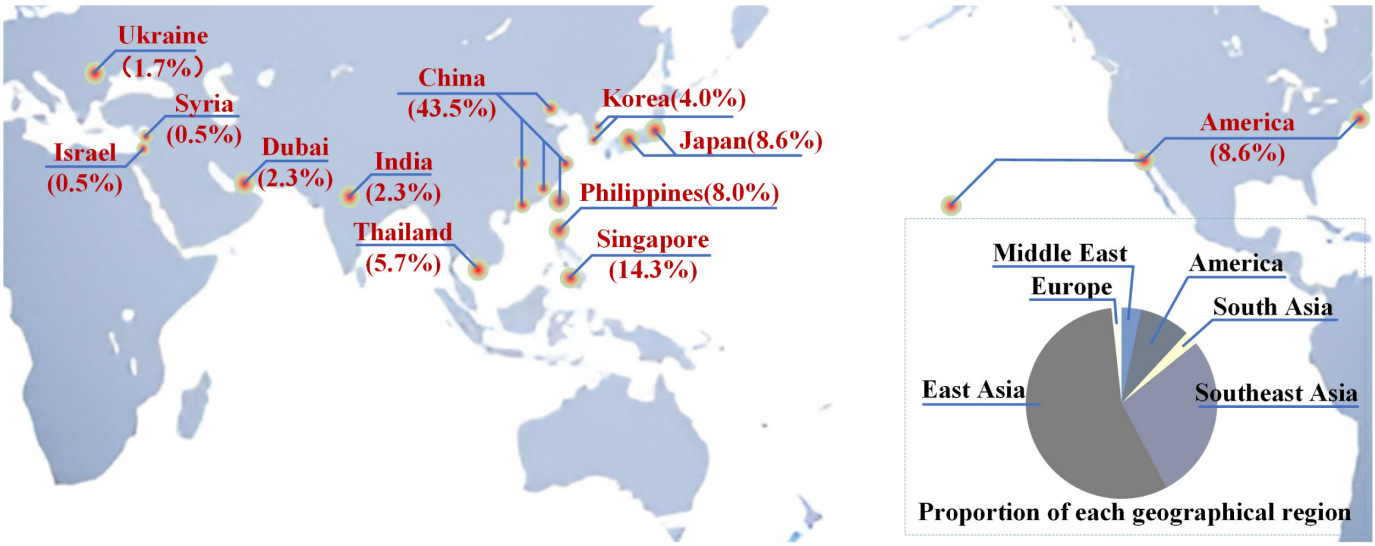


Fig. 4. Geographic distribution of collected SAR images from Gaofen-3.

and interpretation. Our main contributions are summarized as follows.

1) A comprehensive public dataset with enormous quantity, detailed annotations, and attached information based

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on complex-value images has been proposed for the research on SAR trustworthy OD&R tasks. To the best of our knowledge, the FAIR-CSAR dataset is the largest dataset for SAR fine-grained detection and recognition, with more than 340k instances covering 22 subcategories. In addition, it provides more challenging images with some common imaging phenomena, which are closer to the practical application.

2) We benchmark SAR fine-grained OD&R tasks using several state-of-the-art (SOTA) object detectors on the FAIR-CSAR dataset, which can be utilized as the baseline for future work. We believe that this benchmark will promote the development of SAR fine-grained OD&R in remote sensing.
3) A novel MDNet is designed. It applies a specialized pipeline for online enhancement and adds another backbone to acquire the distinguishing information underlying the TF domain (TFD). Contributed to the advantage of the SLC SAR dataset, MDNet achieves superior performance. We hope that it will inspire further applications and research on the FAIR-CSAR dataset.

The rest of this article is organized as follows. Section II reviews mainstream datasets for OD&R. Section III describes the proposed FAIR-CSAR dataset in detail. Section IV conducts experiments and evaluations on the proposed FAIR-CSAR dataset and introduces a novel algorithm suitable for SLC SAR images. Section V finally concludes this article.

II. Related Work

Almost all OD&R algorithms based on deep learning are data-driven, relying on a sufficient number of instances with sufficient granularity to contribute to the training achieving accurate recognition with generalized application. Therefore, a dataset with a larger quantity and finer granularity is essential to the relevant research in specific fields, such as ground-level natural applications and remote sensing applications [17], [37]. Recently, with well-developed large-quantity fine-grained datasets in the field of optical remote sensing applications, such as DOTA [16] and FAIR1M [17], many significant developments of optical OD&D algorithms appear continuously, including RoI Transformer [38], Gaussian Wasserstein distance (loss) (GWD) [39], the RingMo foundation model [40], [41], [42], [43], and so on. However, due to the difficulty in distinguishing the fragmented variable features and unique imaging phenomena, existing SAR datasets struggle to achieve the same quantity as optical datasets and fail in the fine-grained annotation with SAR characteristics [22], [44], [45], [46], [47]. Moreover, the significant gaps between SAR images and optical images make most existing algorithms less effective on SAR images. Therefore, it is urgent to boost the development of relevant algorithms through the construction of the SAR OD&R dataset, including larger quantity instances, finer grained annotation, and more expression on SAR unique characteristics.

A. Datasets for Ground-Level Natural Images

Thanks to the public release of large-scale annotated datasets [28], [29], [37], significant progress in OD&R has been made in ground-level natural applications. The Pascal VOC dataset is one of the exemplary benchmark datasets, which was first proposed in 2005 and gradually established, encompassing various editions with the visual object classes (PASCAL VOC) 2007 and 2012 [28], [29]. The Pascal VOC 2012 dataset, the most widely used among them, contains 11.5k images with annotations covering 20 categories for various tasks, such as image classification, object detection, and image segmentation. As a landmark in computer vision, the Pascal VOC datasets have profoundly fostered the development of OD&R algorithms, such as faster region-based convolutional neural network (R-CNN) and you only look once (YOLO), which are respectively acknowledged as the milestones for the two- and single-stage object detection algorithms. Another famous fundamental dataset is proposed in 2014, namely, microsoft common objects in context (MSCOCO), including more than 80 categories and 896k instances from more than 200k slices [37]. Contributed to the inclusion of more diversified and rich scenes, the MSCOCO dataset provides a more challenging testing platform for researchers.

With high-quality annotations and abundant instances, these datasets have given rise to many classical paradigm algorithms for OD&R, effectively promoting the development of computer vision. Their tremendous success demonstrates that high-quality large-quantity datasets are essential for the breakthroughs of related research.

B. Datasets for Optical Remote Sensing Images

Recently, some specialized datasets have been proposed to advance the development of OD&R in optical remote sensing. To address the limitation of datasets with small scale and insufficient diversity, the detection in optical remote sensing images (DIOR) dataset is proposed, which consists of 192 472 instances from 23 463 images/slices [48]. It includes 20 object categories with high interclass similarity and intraclass diversity. DIOR is annotated with a horizontal bounding box (HBB) and, thus, cannot precisely enclose the rotating objects with an imbalance aspect ratio. To tackle this problem, OBB is introduced, enclosing objects more accurately by considering their orientation. DOTA, the famous dataset with OBB annotation, collects 2806 aerial images with 15 categories of common objects, containing more than 188k instances. DOTA, widely applied in OD&R research, not only drives the development of oriented object detection algorithms [49], [50], [51] but also supports the construction of the MMRotate toolkit [52]. As the benchmark integrating the mainstream-oriented object detection algorithms, MMRotate specifically provides the data processing pipeline compatible with the DOTA dataset, providing great support for the validation of related algorithms based on the DOTA dataset [16]. The FAIR1M dataset is another famous benchmark dataset, with the maximum number of fine-grained instances for high-resolution remote sensing imagery [17]. The FAIR1M consists of more than one million instances from more than 40 000 images/slices, with respect to five categories and 37 subcategories annotated by OBB. The FAIR1M dataset is collected from richer and more diverse optical remote sensing images from different platforms across countries and regions,

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meanwhile containing rich geographic information such as latitude, longitude, and resolution [17]. With its unique advantage of the enormous quantity, FAIR1M provides effective support for the pretraining of RingMo [40] and the relevant fine-tuning of downstream tasks [41], [42].

Datasets for optical remote sensing images have matured as the number of instances, variety of categories, and richness of scenarios reach a higher level. However, due to the unique diverse imaging properties and relatively high difficulty of annotation, the construction of SAR remote sensing datasets is relatively lagging behind.

C. Datasets for SAR Remote Sensing Images

There are many challenges occurring in the production of SAR remote sensing datasets, such as large-quantity fine-grained annotation and detailed attached information. Some SAR remote sensing datasets are constantly being proposed to solve them and provide an experimental basis for object detection algorithms. The mainstream SAR datasets for OD&R are listed in Table I, where their detailed information such as annotation granularity and data volume is also provided.

Due to the remarkable interpretation difficulties and annotation costs, the existing SAR datasets are annotated for easily distinguishable major categories, such as ship and bridge. Most of them contain only a single major category, such as OpenSARShip-1.0 [44], SSDD [19], and AIR-SARShip-1.0 [45]. Others usually roughly categorize various objects by usage, such as MSAR-1.0 [53] and SARDet-100K [54]. Only a few datasets have undergone fine-grained annotation, such as SRSDD-v1.0 [55] and SAR-AIRcraft-1.0 [20].

Datasets with a single category usually aim to support SAR object detection and have made significant contributions to foundational research [56], [57].

1) Among them, datasets with less than 10 000 images/slices are quite common, such as SAD [21], SSDD [19], AIR-SARShip-1.0 [45], high-resolution SAR images dataset (HRSID) [22], and rotated ship detection dataset in SAR images (RSDD-SAR) [47], where SAD is a benchmark dataset for airport detection including 624 images/slices, and all of the others mainly focus on the ship detection. The SSDD, AIR-SARShip-1.0, and HRSID are all annotated with HBB, while RSDD-SAR is annotated with OBB, which provides more accurate spatial information for objects with an imbalance aspect ratio. All of these small-quantity datasets can effectively validate the detection performance of a single category in specific scenarios but struggle to support the validation of generalization applications for the limitation in the number of slices and instances.
2) Both SAR-Ship-Dataset [46] and OpenSARShip-1.0 [44], respectively, contain 43 819 and 11 346 images/slices with larger data quantity. SAR-Ship-Dataset only provides the real-valued intensity images, which can provide visual features but lose other information, such as phase information, included in SLC images. OpenSARShip-1.0 provides both intensity images and SLC images, which can support the research on unique feature extraction and detection technologies in diverse transform domains.

These datasets annotated for a single category make great contributions to the fundamental research on object detection but fail in the further research on object recognition.

Besides datasets that only focus on a single major category, there are also some outstanding SAR datasets that roughly categorize the objects by usage. The MSAR-1.0 is a large-quantity SAR dataset with multiple categories including aircraft, oil tank, bridge, and ship. It comprises 28 448 images/slices from different scenarios, with 60 396 instances including 1851 bridges, 39 858 ships, 12 319 oil tanks, and 6368 aircraft. To further tackle the challenge of limited data resources, SARDet-100K is proposed with a total of 116 598 images/slices and 245 653 instances in six categories including aircraft, ship, car, bridge, tank, and harbor. These datasets for SAR OD&R provide sufficient instances and rough-grained categories, which makes them useful for the improvement of the generalization performance on OD&R. However, these two datasets lack fine-grained annotation for subcategories and only provide intensity images in the real domain, which fail in the validation of fine-grained recognition and are also not applicable for SLC images.

The fine-grained SAR OD&R task is more refined research, aiming at distinguishing various subcategories within the same major category. The SRSDD-v1.0 dataset is proposed to be applied for fine-grained ship detection and recognition on high-resolution SAR images, with OBB annotations suitable for rotating object detection. It contains 666 images with 1024×1024 pixels and 2884 ship instances distributed among six subcategories. Another fine-grained dataset is SAR-AIRcraft-1.0, comprising 4368 images/slices from Gaofen-3 with 16 463 instances annotated by HBB, covering seven subcategories, namely, A220, A320/321, A330, ARJ21, Boeing737, Boeing787, and other. These datasets boast detailed granular category annotations, making them significantly meaningful for fine-grained OD&R. However, all of them are only for a single major category (aircraft or ships), and their images have been meticulously selected and processed, which results in the lack of challenging data and diverse scenarios.

Up until now, SAR datasets have evolved from small quantity, single category, and simple scenes toward large quantity, multiple categories, and diverse scenarios. Considering the unique characteristics and advantages of SAR data, it is necessary to delve deeper into the diverse SAR features varying on imaging conditions by means of the construction of a complete and more challenging dataset. However, there remains a lack of a dataset to address all the aforementioned limitations, which not only possesses the substantial volume, fine-grained category annotations, various scenes, and diverse data domains but also fully validates the unique advantages and characteristics of SAR data with appropriate levels of difficulty and challenge. Such a dataset is expected to not only expand the frontier of current SAR image processing and analysis but also significantly enhance the capability and robustness of OD&R. This is precisely the motivation for our proposed FAIR-CSAR dataset, aimed at filling the gaps in current studies by building a more challenging, widely applicable OD&R dataset for SAR remote sensing images.

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TABLE I
COMPARISON BETWEEN FAIR-CSAR AND OTHER DATASETS FOR SAR REMOTE SENSING IMAGES

Datasets	Images/ Slices	Instances	Resolution	Polarization	Annotation	Data Domain	Categories	Sub categories	Fine-grained
OpenSARShip-1.0	11,346	11,346	2.7m~22m	Dual	HBB	Real, Complex	1	--	N
SSDD	1,160	2,456	1m~15m	Single	HBB	Real	1	--	N
AIR-SARShip-1.0	31	461	1m~3m	Single	HBB	Real	1	--	N
SAR-Ship-Dataset	43,819	59,535	1.7m~25m	Single、Dual、Full	HBB	Real	1	--	N
HRSID	5,604	16,951	0.5m~3m	Dual	HBB	Real	1	--	N
RSDD-SAR	7,000	10,263	2~20m	Single	OBB	Real	1	--	N
SAR Airport Dataset (SAD)	624	624	10m	Single	HBB	Real	1	--	N
MSAR-1.0	28,449	60,396	≤1m	Single	HBB	Real	4	--	N
SRSDD-v1.0	666	2,884	1m	Single	OBB	Real	1	6	Y
SAR-AIRcraft-1.0	4,368	16,463	1m	Single	HBB	Real	1	7	Y
SARDet-100K	116,598	245,653	0.1m~25m	Single	HBB	Real	6	--	N
FAIR-CSAR	**106,672**	**349,002**	**1m~5m**	**Single, Dual**	**OBB**	**Real, Complex**	**5**	**22**	**Y**

III. DETAILS OF THE FAIR-CSAR DATASET

A. Image Collection and Preprocessing

1) Data Collection: The visual features of SAR generally exhibit significant variations across different scenes, imaging conditions, and modes [23], [58]. To ensure that the OD&R algorithms based on the FAIR-CSAR dataset can achieve satisfactory generalization performance, this dataset encompasses abundant scenes and collects data under diverse imaging conditions and modes.

This dataset comprises 175 scenes of Gaofen-3 SLC images collected from 32 different airports, ports, rivers, and cities. The geographic distribution is illustrated in Fig. 4. The extensive geographical distribution simultaneously ensures the diversity of imaging conditions such as incidence angles. In addition, this dataset enhances the diversity of imaging modes from the perspectives of both the acquisition mode and the polarization mode. The dataset is divided into two parts based on the acquisition mode: the spotlight (SL) dataset and the fine strip-map (FSI) dataset.

The former includes 167 754 instances across five major categories and 22 subcategories, while the latter comprises 180 970 instances within three categories. For the polarization mode, the SL dataset consists of single-polarization data, covering the HH, HV, VH, and VV polarization modes, while the FSI dataset collects images under the dual-polarization mode.

2) Data Preprocessing: The data preprocessing primarily includes coarse geometric correction and basic quantification. Different from real-valued intensity images, the SLC SAR images present the slant range rather than the ground projection distance, and the pixel resolutions along the range and azimuth are usually inconsistent [59], [60]. This issue results in obvious geometric distortions of targets, as shown in Fig. 5(a). To enhance the visual features, the coarse geometric correction is first applied to the slices according to γ, the ratio of ground resolution along the azimuth and range, which is given as follows:

$$\gamma = \mathrm{Res}_a \cdot \sin\theta_{\mathrm{in}} / \mathrm{Res}_r \tag{1}$$

where Res_a and Res_r, respectively, denote the pixel resolution along azimuth and range and θ_{in} denotes the incident angle of the electromagnetic wave. Since the azimuth antenna pattern (AAP) in the SL mode varies with spatial position [61], [62], the rough geometric correction was performed on the local slices. As shown in Fig. 6, the process of rough geometric correction is given as follows: the Fourier transform is performed on the SLC slice along the azimuth to obtain its range-Doppler spectrum, and the Doppler center frequency is estimated [63]. When the ratio γ is less than 1, the range-Doppler spectrum is truncated along azimuth centered on the Doppler center frequency according to the ratio γ; when the ratio γ is greater than 1, the range-Doppler spectrum is zero-padded in the high-frequency area along the azimuth. The inverse Fourier transform is then performed to obtain the roughly geometrically corrected slice. These geometrically corrected slices form the quintessential part of this dataset. Fig. 5(b) and (d) shows the range-Doppler spectrum before and after the truncation. Fig. 5(c) displays the image after the geometric correction.

After the geometrical correction, an adaptive quantization method is applied for the SLC SAR images to obtain the intensity images. The dataset contains SAR images/slices of both complex-valued format and real-valued format. In addition, the adaptive quantization method will be detailed in Section IV.

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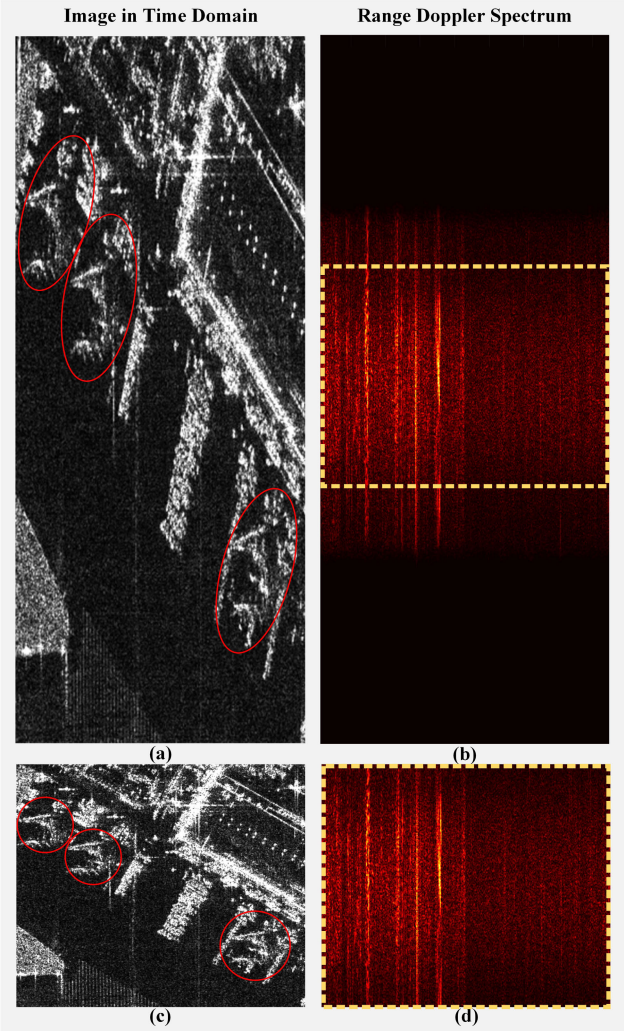


Fig. 5. Geometric distortion and correction processing. (a) and (b) display the original image and range Doppler spectrum before the geometric correction, respectively. (c) and (d) display the image and range Doppler spectrum after the geometric correction, respectively.

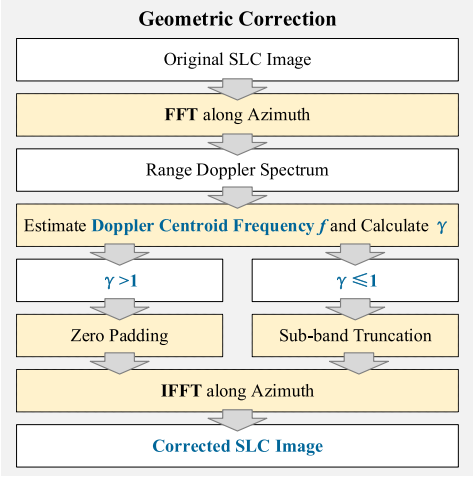


Fig. 6. Schematic of the geometric correction process.

B. Category Design and Statistical Characteristics

1) Category Deign: Different from optical images, SAR images have larger intraclass variability among scattering feature space, coupled with a closer interclass gap [64]. This results in more difficulties occurring in SAR object classification, especially fine-grained recognition. Therefore, the vast majority of SAR datasets were made for easily identifiable targets with larger sizes, such as aircraft and ships. Also, the category design was limited to these major categories. They lack fine-grained annotation information, which cannot support fine-grained recognition applications. In addition, the abovementioned major categories are also susceptible to feature confusion from other static targets such as tower cranes, oil tanks, and bridges, especially in recognition applications with wide-swath scenarios.

TABLE II
CATEGORY DESIGN OF THE FAIR-CSAR DATASET

Categories	Subcategories
Aircraft	Airbus A220, Airbus A320, Airbus A330, Boeing 737, Boeing 747, Boeing 767, Boeing 777, Gulfstream, Fokker-50, Helicopter, Airfreighter, Other Aircraft
Ship	Bulk Carrier, Oil Tanker, Container Ship, General Cargo Ship, Warship, Motion Defocusing Ship, Other Ships
Oil Tank	Oil Tank
Bridge	Bridge
Tower Crane	Tower Crane

To meet the demand for challenging fine-grained recognition, the proposed FAIR-CSAR dataset is divided into five major categories, that is, aircraft, ship, oil tank, bridge, and tower crane. Based on these, a detailed fine-grained category design is developed, comprising 22 subcategories (as shown in Table II). The details are presented as follows.

The major category of aircraft is subdivided into 11 common subcategories, including Airbus A220, Airbus A320, Airbus A330, Boeing 737, Boeing 747, Boeing 767, Boeing 777, Gulfstream, Fokker-50, Helicopter, and Airfreighter. Another major category of ship is subdivided into six common subcategories, including bulk carrier, oil tanker, container ship, general cargo ship, warship, and motion-defocusing ship, where the motion-defocusing ship is recorded as an independent subcategory for its features significantly different from that of other clear ship targets. Furthermore, targets falling without the aforementioned specific subcategories are labeled as "Other Aircraft" and "Other Ship."

In light of various abilities on feature representation between SL and FSI SAR images [65] the annotation categories are designed in differentiated forms. The high-resolution images acquired in SL mode are labeled in five major categories and 22 subcategories. However, the medium-resolution images from the FSI mode are annotated with three categories, i.e., ship, bridge, and oil tank. All of the subcategories are presented in the schematic form, as shown in Fig. 7.

Our category design offers richer and more detailed annotation information, which includes other major categories easily confused with aircraft and ships. Some categories with visual recognition challenges are also included. All of these are to meet the demand used for diverse, fine-grained recognition applications.

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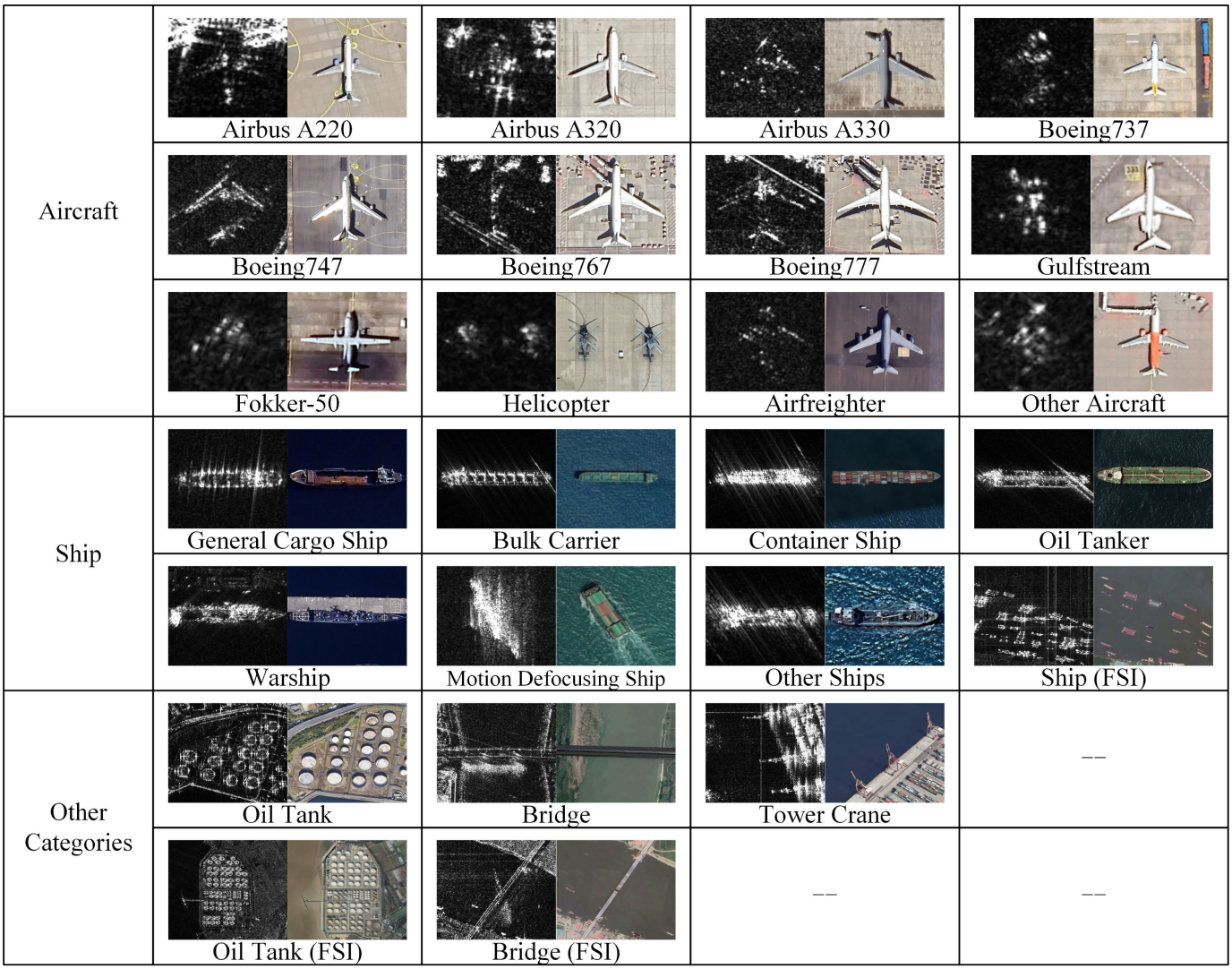


Fig. 7. Visualization of all the five major categories and 22 subcategories within the FAIR-CSAR dataset and corresponding optical remote sensing images for reference.

2) Statistical Characteristics: The basic statistical characteristics of each category are provided in Fig. 8(a)–(c).

1) The number of instances in each category. It is displayed in Fig. 8(a). The SL part of FAIR-CSAR presents a significant disparity in the number of instances for some categories, with the maximum instance ratio reaching as high as 49.5:1 for "Oil Tank" and "Boeing747." The FSI part has a relatively balanced distribution across different categories, but there is still variation with the maximum instance ratio reaching 14.8:1. The issue of category imbalance is one of the major challenges for OD&R tasks as it may lead to poor recognition performance on categories with small-quantity instances.
2) The average scale of OBB for each category. This statistical indicator is calculated based on OBB annotations. It is presented in Fig. 8(b), which also includes the corresponding standard deviation. The square root of the average area, which is used as the measure for the average scale, varies with subcategories. The maximum ratio of 6.6:1 is achieved between "Bulk Carrier" and "Helicopter" subcategories. It reveals that both large- and small-scale subcategories are included in FAIR-CSAR. In addition, some subcategories, such as "Bridge," exhibit remarkable standard deviation. Its large intraclass variance appearing in scale further challenges the detectors to distinguish objects with various scales. All of these reveal that FAIR-CSAR contributes to the validation of the performance on multiscale OD&R.
3) The average aspect ratio of OBB for each category. It is given in Fig. 8(c), where the dashed line is provided as a reference for the aspect ratio equaling 1. Some categories, such as "Ship," "Bridge," and "Tower Crane," exhibit narrow shapes with imbalanced aspect ratios, whereas other categories appear square-like. It reveals that FAIR-CSAR covering categories with diverse aspect ratios is also challenging for accurate oriented estimation and object recognition.

C. Image Annotation

1) Annotation Format: The annotation information provided by the FAIR-CSAR dataset includes OBB and category labels, along with the incidence and attitude angles, and

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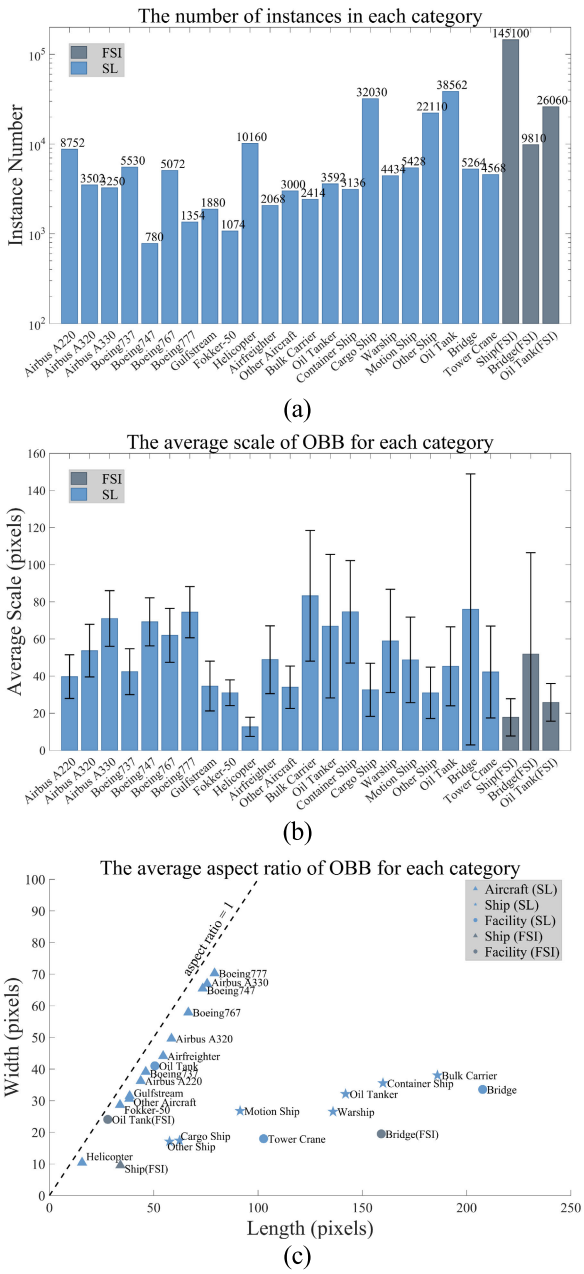


Fig. 8. Statistical characteristics of each category in FAIR-CSAR. (a) Instance number in each category. (b) Average scale of OBB for each category, along with the standard deviation. (c) Average aspect ratio of the OBB for each category. "Motion Defocusing Ship" and "General Cargo Ship" are abbreviated as "Motion Ship" and "Cargo Ship" in this figure, respectively.

other basic imaging parameters. Both DOTA and Pascal VOC annotation formats are provided.

The HBB is the cost-effective annotation format used by most existing datasets but provides limited spatial information about the objects [66], [67]. For objects with a significant disparity in aspect ratio, HBB fails to capture the true spatial information of the target, covers redundant background information, and even overlaps with other targets, resulting in a significant increase in intraclass variance. OBB better conforms to the target and precisely reflects the orientation and spatial information of the target. To get more accurate position

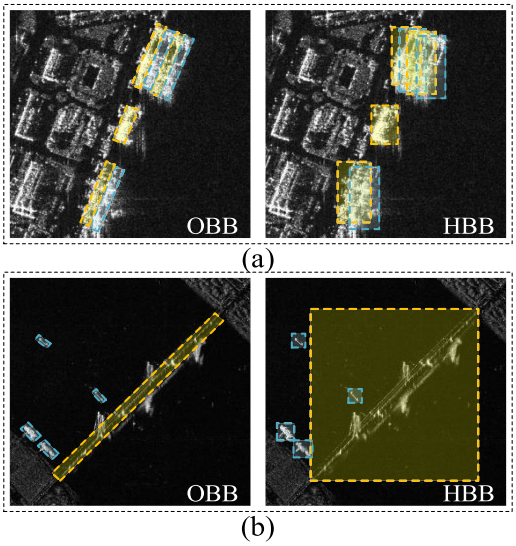


Fig. 9. Comparison between OBB and HBB. (a) and (b) compare the OBB and HBB annotations in different situations.

and shape information, the FAIR-CSAR dataset is annotated with OBB. Fig. 9 compares the two annotation formats.

The radar cross section (RCS) is sensitive to the angle between the electromagnetic wave and the target orientation, which influences the scattering features of the target and takes latent research value [68]. Therefore, the FAIR-CSAR dataset provides the incidence and attitude angles for future potential studies, where the target attitude angle is additionally manually labeled. Fig. 10 displays the concrete geometric depiction of the incidence and attitude angles. In addition, to facilitate the data processing and enhancement for researchers engaged with the FAIR-CSAR dataset, the FAIR-CSAR dataset provides the basic imaging parameters for each image, such as the pulse repetition frequency (PRF) and center frequency.

In addition, the DOTA format annotation files only provide the OBB and category annotations, while the Pascal VOC format annotation files preserve all the mentioned annotation information in extensible markup language (XML) files, as shown in Fig. 11.

2) Annotation Quality: To guarantee the quality of the annotations, the annotation process was conducted under a complete and strict quality control protocol. In the first stage, we performed corrections and visualizations on the SLC data. We invited experts in the SAR field to summarize the characteristics of each category and form an annotation guideline. In the second stage, annotators were divided into pairs, referencing the annotation guideline and corresponding optical images, to independently annotate the SAR images. They then examined each other's annotations and merged them to achieve more accurate annotations. Subsequently, a supervisor was invited to further review the quality of the annotations, encompassing the positions of the bounding boxes, categories, and orientations. Finally, experts in the SAR field were enlisted to conduct the quality inspection of the annotations.

D. Characteristics and Challenges of Dataset

As illustrated in Section II, existing SAR datasets for fine-grained OD&R have several common limitations, including

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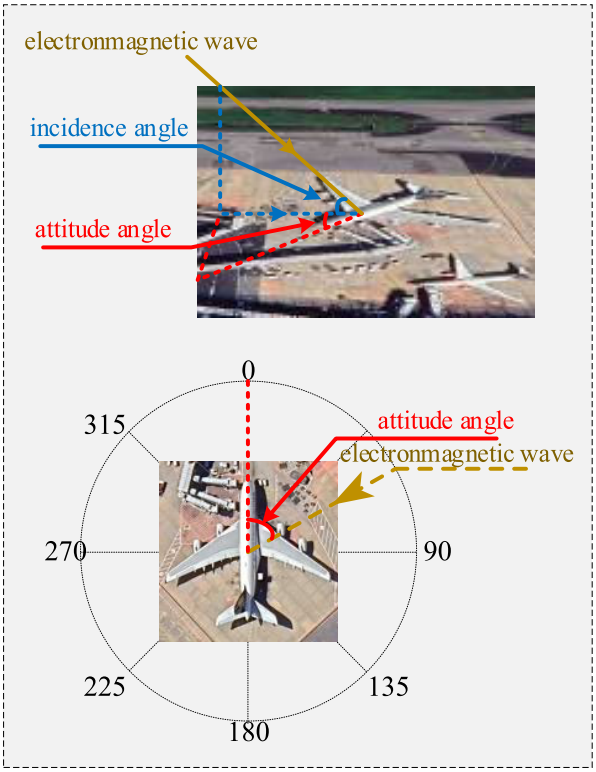


Fig. 10. Concrete geometric depiction of the incidence and attitude angles. The upper part of the illustration represents the 3-D geometric perspective, whereas the lower part provides the vertical view. It is worth mentioning that the target attitude angle in FAIR-CSAR is additionally manually labeled, which needs more expertise and effort.

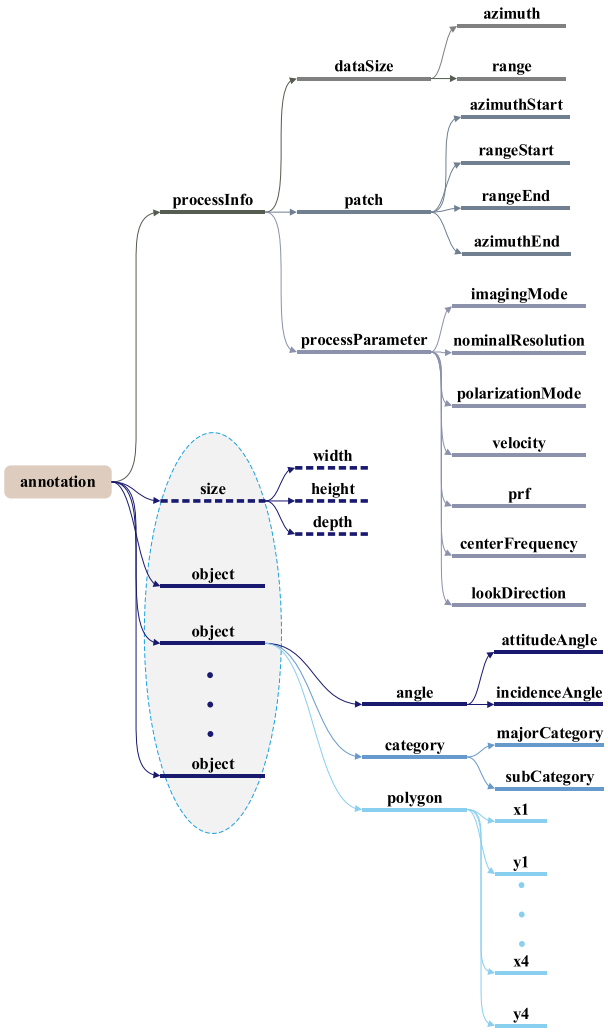


Fig. 11. Annotation information of FAIR-CSAR recorded in XML files.

insufficient quantity, coarse-grained annotation, singular data pattern and similar collection scenario, and lack of challenges for the actual application. To break these limitations and contribute to the promotion of OD&R performance, the FAIR-CSAR dataset is proposed, with more than 340k instances and 22 fine-grained categories. The diversities and challenges of FAIR-CSAR are summarized as follows.

1) Massive Images/Slices With Annotation Instances: FAIR-CSAR is the largest dataset for OD&R in the SAR imagery so far. It encompasses 175 scenes of data collected from 32 cities worldwide, offering over 100k data slices and more than 340k instances from images acquired in multimode (namely, the acquisition mode, the polarization mode, and the data storage mode). Sufficient images/slices and instances are essential to the validation of various OD&R algorithms under diverse conditions and can effectively support the research of OD&R in SAR imagery.

2) Fine-Grained Categories Annotation and Attached Information: FAIR-CSAR provides annotations for 22 fine-grained subcategories across five major categories. Other important details, namely, the incidence and attitude angles, are also attached in the annotation file, which is helpful to the performance validation under various observed angles. In addition, crucial imaging processing parameters, such as radar center frequency and PRF, are also included, which offers substantial support for data transformation, enhancement, object detection, and fine-grained recognition.

3) Diversified Collection Scenarios and Data Patterns: FAIR-CSAR includes large-quantity images/slices, collected from various conditions and diverse scenarios, including rivers, oceans, nearshore docks, airports, towns, villages, islands, and so on. For example, in the scenario of oceans, the annotation data include high-quality images, as well as images subjected to obvious sea clutter. The abundance of scenarios provides the basis for the generalization performance improvement of detectors.

In addition, FAIR-CSAR provides two data storage modes for selection. One is the intensity images/slices in the real domain; the other is the SLC images/slices in the complex domain. The former is commonly used for mainstream detectors and directly provides the visual features. More importantly, the latter that contains abundant spectral and imaging information can provide extra effective information for OD&R in SAR imagery. For acquisition mode, the FAIR-CSAR dataset covers both the SL mode and the FSI mode of the Gaofen-3 satellite [65]. For the polarization mode, the FAIR-CSAR dataset includes single polarization and dual

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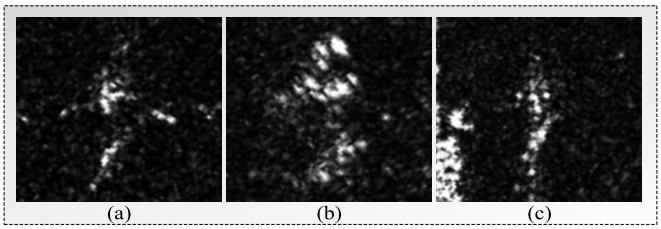


Fig. 12. "Boeing 737" appearing differences under various imaging conditions. (a)–(c) Visual features of "Boeing 737" under different imaging conditions.

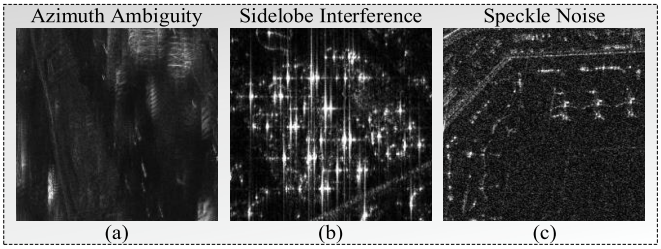


Fig. 13. Common image quality problems in the SAR images. (a) Serious azimuth ambiguities that occlude the ships on the river. (b) Sidelobes that destroy the visual features of the oil tank. (c) Serious speckle noise drowns out the aircraft targets.

polarization. The diverse data patterns endow FAIR-CSAR with universal research value.

4) Research-Worthy Challenges for Actual Application: The research-worthy challenges contained in FAIR-CSAR are given as follows.

1) Large intraclass variance: The target RCS, sensitive to the imaging conditions, results in that the visual feature of the same category appears significant difference. The remarkable intraclass variance makes the OD&R task more difficult. As shown in Fig. 12, "Boeing 737" presents significant differences in the visual features under various imaging conditions.
2) Common image quality problems: They are quite common in SAR imagery, such as azimuth ambiguity [69], sidelobe interference [70], and speckle noise [71]. As shown in Fig. 13, these problems tend to cause direct damage to visual features, resulting in false alarms and missed detection.
3) Motion defocusing: Since SAR imaging relies on precise phase compensation to achieve high-resolution imaging along azimuth, the phase offsets caused by target motion will result in defocusing visual features, further leading to miss detection [72].
4) Multipath effect: Multipath effect refers to the phenomenon that electromagnetic waves reflect off multiple surfaces before returning to the sensor, causing distortions and artifacts in the images [73]. The multipath effect interferes with the visual features and tends to result in false alarms. Fig. 14 presents the multipath effects on aircraft wings and bridges.
5) Incremental information extraction: SAR data contain valuable information in the time domain (TD), the frequency domain, and other transformation domains. The extra information can be extracted from SLC images and utilized to distinguish different fine-grained categories, which is further applied to the improvement of performance on OD&R. However, current datasets and studies on incremental information extraction are insufficient.

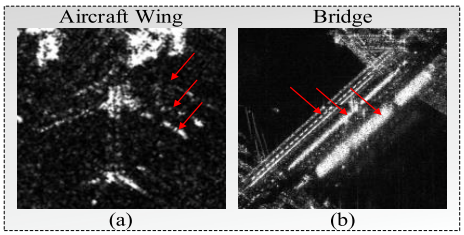


Fig. 14. Multipath effects on aircraft wings and bridge. (a) shows the multipath effect on aircraft wings, while (b) shows the multipath effect on the bridge.

Most of these challenges weaken the performance of OD&R. It is worthwhile to study and solve these issues for SAR actual application with the help of FAIR-CSAR.

E. Dataset Splits

The proposed dataset is divided into the training and validation sets (denoted as the trainval set) and the test set. To validate the generalization performance of various detectors, the test set encompasses images/slices not only in different time phases but also acquired from scenes that have never appeared in the trainval set. Unfamiliar scenes present a great challenge to the generalization capabilities of detectors.

The trainval set of the SL dataset is derived from 105 scenes of SLC SAR images, while the test set is derived from other 26 scenes. The trainval set of the FSI dataset is derived from 34 scenes, and the test set is derived from the other ten scenes. Both the images/slices and annotation files of the trainval set are provided in FAIR-CSAR. For the test set, only the images/slices are available, but researchers can check the detection results via the provided benchmark evaluation server.

IV. Algorithm Analysis

The oriented OD&R experiments based on representative detectors are conducted on FAIR-CSAR to confirm the challenges and advantages of the proposed dataset. In addition, an MDNet with a dual-backbone structure is proposed as a novel framework to mine detailed information underlying SLC SAR images. MDNet applies a specialized pipeline for online enhancement and adds another backbone to acquire the distinguishing information underlying the TFD with the help of short-time Fourier transformation (STFT). Contributed to the advantage of the SLC SAR datasets, MDNet achieves the SOTA performance on OD&R.

A. Evaluation Metrics

To evaluate the performance of various detectors, the $F1$ score and the mean average precision (mAP) are used for assessments and analyses [74], [75].

$F1$ score comprehensively reflects the recall and accuracy and can be calculated as

$$F1 = \frac{2 \cdot P_r \cdot R_c}{P_r + R_c} \tag{2}$$

where P_r and R_c are denoted as precision and recall, respectively.

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TABLE III
BENCHMARK RESULTS OF THE FAIR-CSAR DATASET ON THE SL MODE

Category	RetinaNet [77]		Faster-RCNN [30]		RoI Transformer [38]		GWD [39]	
	F1	AP (%)	F1	AP (%)	F1	AP (%)	F1	AP (%)
Airbus A220	0.58	55.8	0.51	47.4	0.54	47.0	0.55	58.4
Airbus A320	0.44	30.6	0.34	27.4	0.40	29.0	0.50	38.3
Airbus A330	0.62	57.2	0.64	55.5	0.66	66.0	0.63	62.3
Boeing737	0.50	46.5	0.50	44.6	0.52	48.3	0.42	38.3
Boeing747	0.46	34.4	0.69	64.8	0.69	60.3	0.59	55.6
Boeing767	0.52	48.4	0.54	36.8	0.58	47.4	0.51	48.0
Boeing777	0.32	26.1	0.30	22.6	0.42	23.9	0.30	23.4
GulfStream	0.35	25.1	0.42	22.7	0.37	20.8	0.45	39.3
Fokker-50	0.57	55.8	0.51	52.4	0.58	56.5	0.46	50.1
Helicopter	0.32	19.3	0.38	27.7	0.51	41.8	0.36	24.8
Airfreighter	0.36	29.6	0.52	43.7	0.46	38.9	0.40	31.3
Other Aircraft	0.20	16.9	0.17	5.0	0.21	11.8	0.21	11.8
Bulk Carrier	0.46	42.1	0.69	64.9	0.70	60.4	0.56	50.9
Oil Tanker	0.19	10.1	0.32	26.7	0.31	26.6	0.24	13.5
Container Ship	0.23	13.6	0.39	30.0	0.37	30.9	0.23	15.7
General Cargo Ship	0.38	29.7	0.42	32.1	0.51	41.4	0.41	32.9
Warship	0.13	4.5	0.32	19.0	0.36	25.2	0.18	9.2
Motion Defocusing Ship	0.39	32.0	0.38	29.2	0.38	29.5	0.36	28.9
Other Ships	0.28	22.6	0.35	26.4	0.38	28.2	0.35	27.5
Oil Tank	0.65	56.8	0.64	55.6	0.63	49.2	0.65	57.3
Bridge	0.29	21.5	0.40	25.6	0.42	32.4	0.31	23.4
Tower Crane	0.28	20.1	0.29	16.2	0.33	20.8	0.26	17.7
Average	**0.39**	**31.8**	**0.44**	**35.3**	**0.47**	**38.0**	**0.40**	**34.5**

TABLE IV
BENCHMARK RESULTS OF THE FAIR-CSAR DATASET ON THE FSI MODE

Category	RetinaNet [77]		Faster-RCNN [30]		RoI Transformer [38]		GWD [39]	
	F1	AP (%)	F1	AP (%)	F1	AP (%)	F1	AP (%)
Bridge	0.34	18.2	0.43	32.1	0.48	29.9	0.37	23.3
Oil Tank	0.58	49.3	0.65	53.5	0.65	53.2	0.62	52.1
Ship	0.55	41.0	0.55	43.0	0.60	50.8	0.59	47.2
Average	**0.49**	**36.2**	**0.54**	**42.9**	**0.58**	**44.6**	**0.52**	**40.9**

Precision P_r signifies the accuracy of the positive predictions, and recall R_c aims to measure the performance of capturing all relevant objects in the image. They are calculated as

$$P_r = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{3}$$

$$R_c = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{4}$$

where TP, FP, and FN denote the number of true positive, false positive, and false negative instances, respectively. Here, the intersection over union (IoU) between the predicted bounding box and the ground truth is used as the criterion to distinguish the positive and negative instances. TP denotes the predicted instances with the IoU higher than the preset threshold.

FP is judged by the IoU lower than the preset threshold and considered as a false alarm. In addition, the existing instances missed by detectors are recorded as FN.

mAP is another important indicator, the mean value of average precision (AP) for all categories. AP refers to the area under the precision–recall curve, which is the average of

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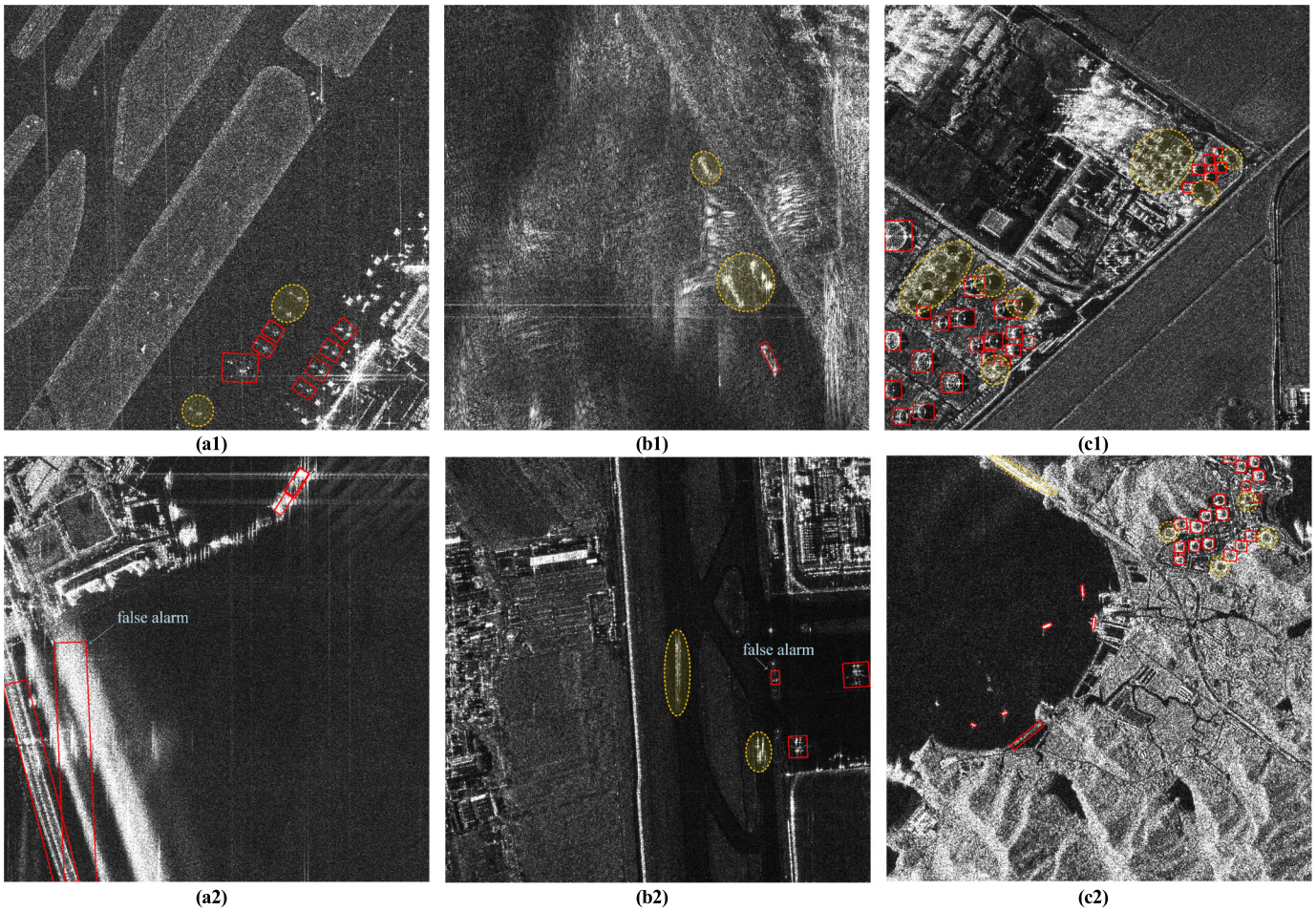


Fig. 15. Visualization of detection results of testing on the FAIR-CSAR dataset using the RoI Transformer. Some typical challenges are (a1) small object disturbed by speckle, (a2) multipath disturbance, (b1) azimuth ambiguities, (b2) motion defocusing, and (c1) and (c2) densely distributed objects in SL and FSI modes. The detection results and the missed targets are enclosed by red solid rectangles and yellow dotted circles, respectively.

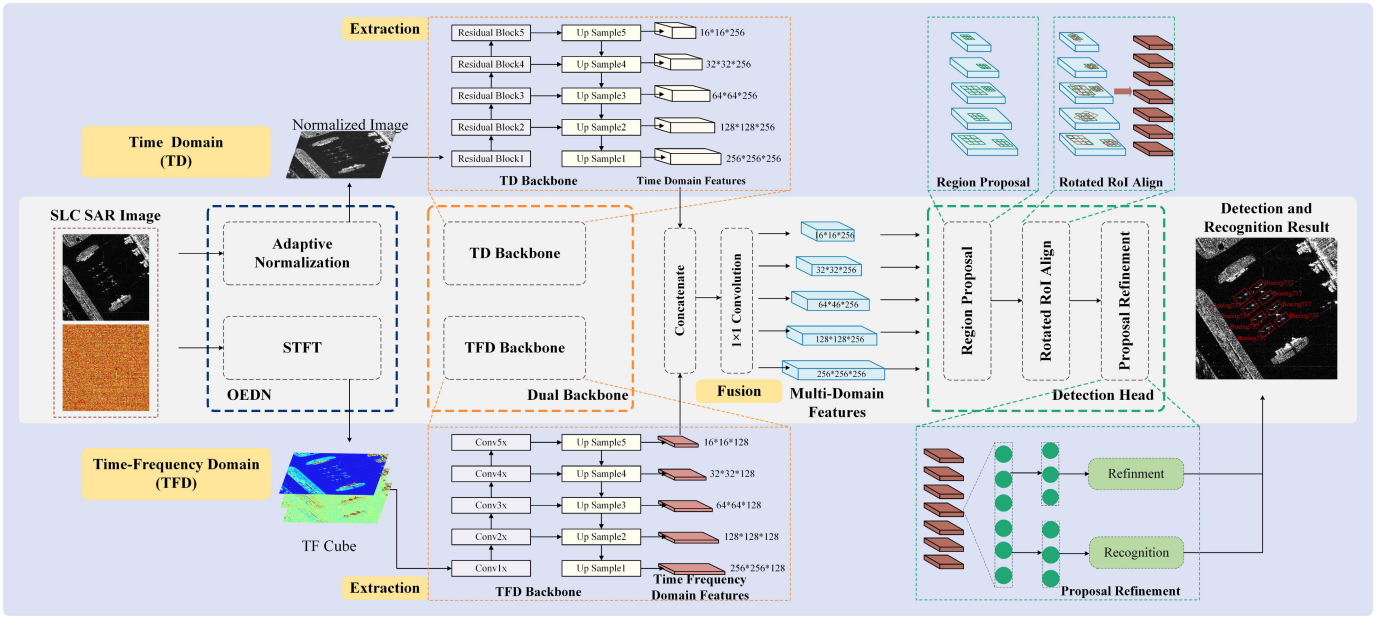


Fig. 16. Framework of MDNet with the dual-backbone structure.

precision values at various recall levels

$$\mathrm{AP} = \int_0^1 P_r(R_c) dR_c. \tag{5}$$

B. Benchmarks

Dataset: Each scene of the SLC image occupies a large storage space, nearly 2 GB, leading to an unacceptable

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TABLE V
DETECTION AND RECOGNITION RESULTS OF THE MDNET ON THE SLC SAR IMAGE ACQUIRED BY THE SL MODE

Category	MDNet		Category	MDNet	
	F1	AP (%)		F1	AP (%)
Airbus A220	0.54	54.7	Other Aircraft	0.23	17.3
Airbus A320	0.37	32.4	Bulk Carrier	0.70	60.5
Airbus A330	0.70	67.8	Oil Tanker	0.28	27.4
Boeing737	0.58	53.2	Container Ship	0.38	31.7
Boeing747	0.70	62.8	General Cargo Ship	0.50	43.9
Boeing767	0.56	49.4	Warship	0.31	24.6
Boeing777	0.27	17.5	Motion Defocusing Ship	0.42	34.4
GulfStream	0.40	26.4	Other Ships	0.38	30.2
Fokker-50	0.63	63.2	Oil Tank	0.68	62.7
Helicopter	0.47	45.7	Bridge	0.38	29.4
Airfreighter	0.39	31.1	Tower Crane	0.38	27.1
Average F1	**0.47**		**Average AP (%)**	**40.6**	

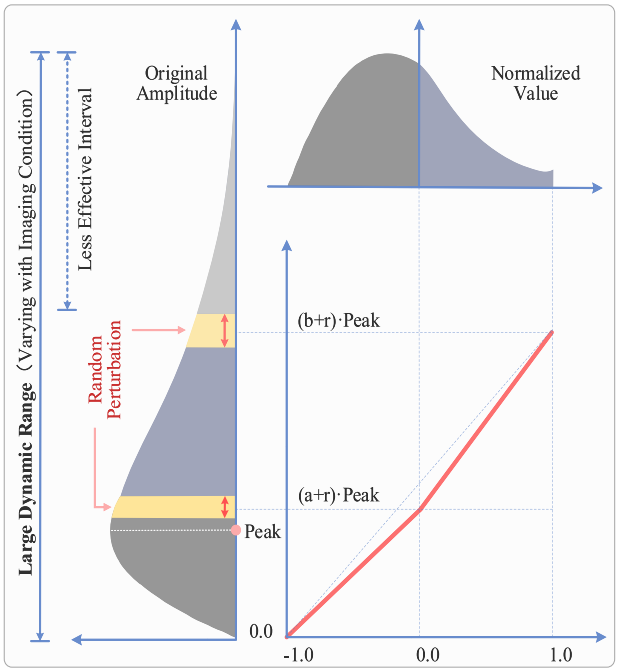


Fig. 17. Illustration of the process of distribution normalization. The mapping principle from the original amplitude to the normalized value is presented.

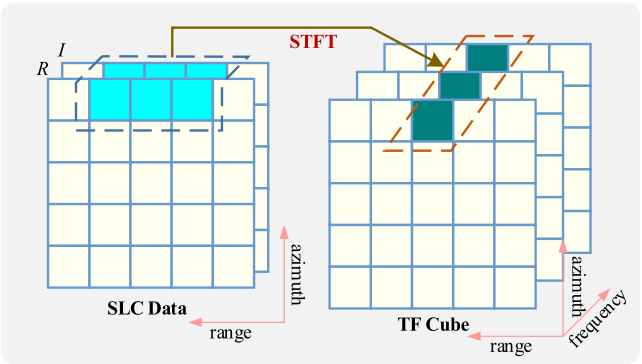


Fig. 18. Generation of TF cube as the input of the TFD backbone.

TABLE VI
DETECTION AND RECOGNITION RESULTS OF THE MDNET ON THE SLC SAR IMAGE ACQUIRED BY THE FSI MODE

Category	MDNet	
	F1	AP (%)
Bridge	0.47	36.9
Oil Tank	0.68	58.5
Ship	0.67	61.2
Average	**0.61**	**52.2**

computational load and storage burden. To ensure that they remain acceptable, each entire scene of the SLC image is cropped into 1024 × 1024 slices with a stride of 512.

Baseline models: With research on the SOTA-oriented object detection algorithms, the most representative algorithms act as the baseline models, including RetinaNet [76], [77], faster R-CNN [30], RoI Transformer [38], and GWD [39] with the ResNet-50 [78] backbone. All of them are implemented based on the code library [52], covering both single- and two-stage algorithms. Also, classical algorithms with enhancement in feature extraction or loss function are included.

Experimental settings: All models, including the following proposed MDNet, were trained under consistent settings. The training process was conducted on the single 2080Ti GPU with the batch size set to 2. The initial learning rate was set

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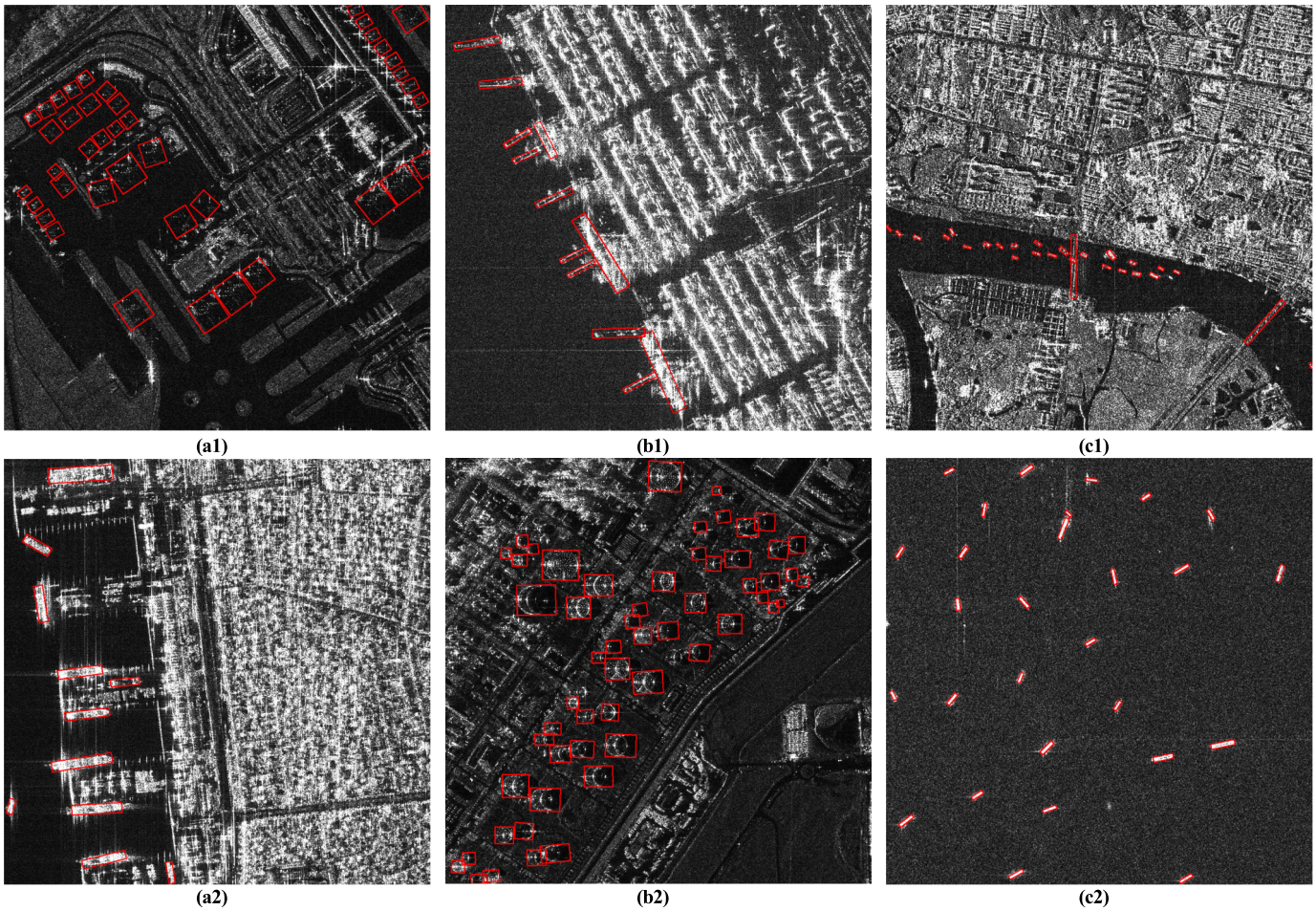


Fig. 19. OD&R results of MDNet. (a1), (a2), (b1), and (b2) Detection results under the SL mode in the scenes of airport, port, and so on. (c1) and (c2) Detection results under the FSI mode in the scenes of river and ocean surface, respectively.

at 0.0025, decaying to 10% every ten epochs. Ten percent of the training set was randomly partitioned as the validation set, and the model with the highest mAP on the validation set was chosen for the test.

Baseline results: Four baseline models are trained and verified on the FAIR-CSAR dataset. The corresponding performance on SAR images acquired by SL and FSI modes is, respectively, displayed in Tables III and IV, where $F1$ score and AP for each category are recorded. These baseline results show the universality of the FAIR-CSAR dataset and present the difficulty of all existing generic detectors applied to fine-grained objects. For categories with sufficient instances and distinct features, all detectors can achieve good performance. For example, the AP of "Oil Tank" reaches approximately 50% or even higher. For some categories with distinct features but few instances, such as "Boeing 777," the corresponding performances are not satisfactory. The instances of these categories are helpful to the verification of imbalanced fine-grained recognition tasks. Moreover, some categories with indistinctive features or severely unbalanced aspect ratios, such as "Tower Crane" or "Bridge," are also difficult to distinguish. Given that the acquisition of salient features from other transform domains is a potential solution, the images/slices including these categories from SLC SAR images are essential to the relevant verification.

Failure cases: Fig. 15 presents the visualization of detection results using the RoI Transformer. In addition to object recognition, there are undetected objects and false alarms in the results, which are caused by some typical challenges in actual application, such as speckle noise, multipath disturbance, azimuth ambiguities, motion defocusing, and densely distributed scenarios.

C. MDNet on FAIR-CSAR

As shown in the above experiment, the performance of existing algorithms on object detection and fine-grained recognition is still insufficient, which results from the fragmented visual features with the large intraclass difference appearing in SAR images of the real domain. Considering the advantage of FAIR-CSAR based on SLC images and detailed attached information, we propose an MDNet, which is adapted for SLC SAR images stored in Uint16 format. Its specialized pipeline can realize online adaptive enhancement and transformation and extract additional TFD information to promote performance. Fig. 16 presents the framework of MDNet.

Framework: MDNet is built on RoI Transformer [38], which applies a specialized pipeline consisting of three stages: online enhancement and distribution normalization (OEDN), the dual backbone for feature extraction and fusion (DBEF),

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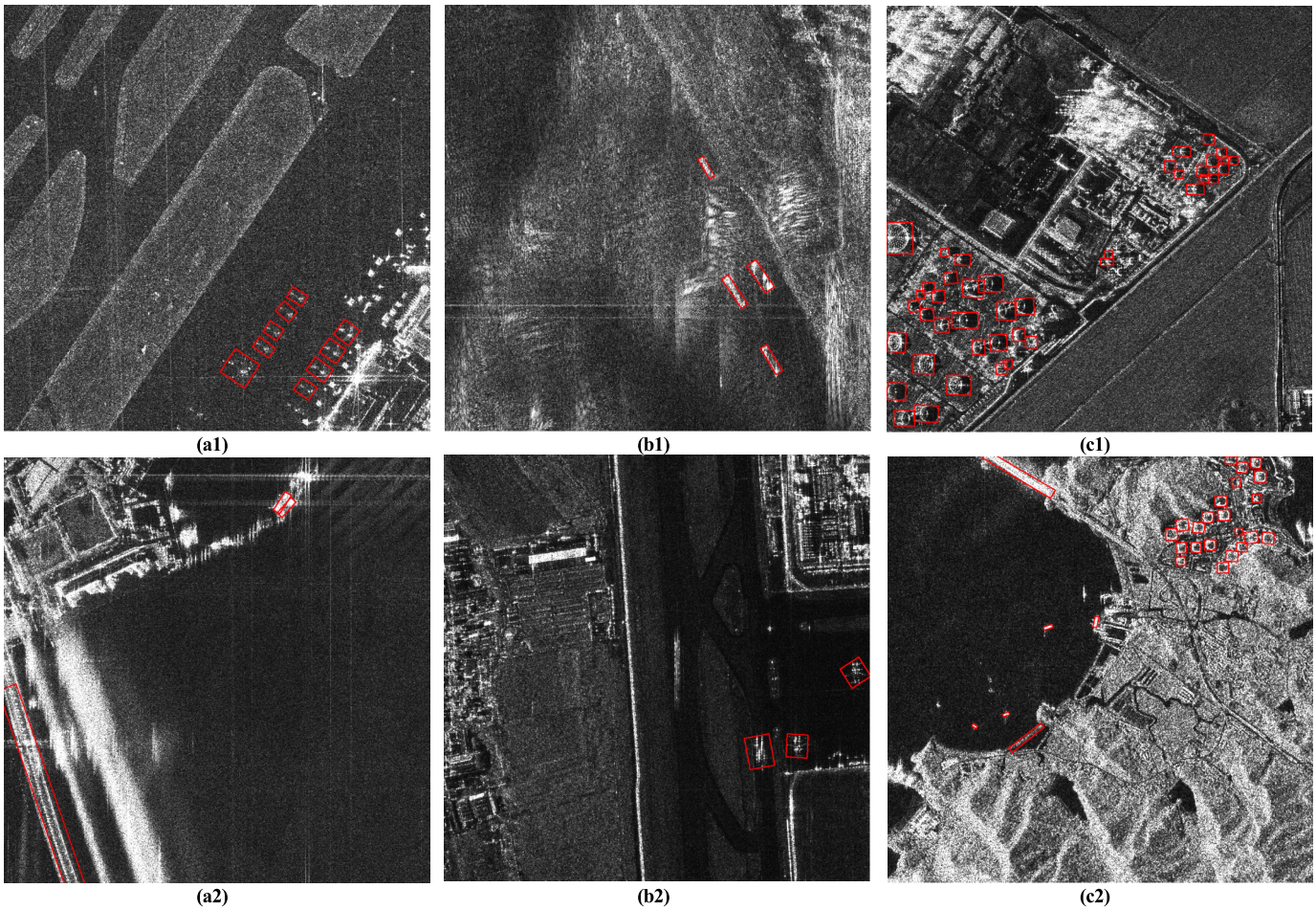


Fig. 20. Visualization of detection results of testing on the difficult cases included in the FAIR-CSAR dataset using MDNet. Some typical challenges are (a1) small object disturbed by speckle, (a2) multipath disturbance, (b1) azimuth ambiguities, (b2) motion defocusing, and (c1) and (c2) densely distributed objects in SL and FSI modes. MDNet performs better than RoI Transformer.

and detection head for OBB regression and recognition. The first stage can not only acquire the effective principal component included in TD with a large dynamic range but also achieve the local spatial correlation information underlying TFD. In the second and third stages, the above acquired information is further extracted by a dual backbone, namely, the TD backbone and the TFD backbone. Their multiscale feature maps are fused via 1×1 convolution and further applied for location regression and classification. Contributed to better features acquired by multidomain extraction and fusion, superior performance on fine-grained oriented OD&R tasks is achieved.

1) Online Enhancement and Distribution Normalization: OEDN is designed to provide the essential inputs for the subsequent dual-backbone structure. It consists of two parts: one is the adaptive normalization module, and the other is the STFT analysis module for the generation of the TF cube.

The adaptive normalization module is designed to reduce the quantization differences in various scenes and imaging qualities and applies limited bits (8 bits) to describe the feature information of the target and background in detail, which effectively compresses the large dynamic range existing in SAR [79]. The effectiveness of this processing is based on the facts as follows: 1) the amplitude of the SAR SLC image (stored in Uint16) appears unimodal long-tailed distribution [80], [81] and 2) most extremely strong-scattering information lying in the tail of amplitude distribution is less effective; if it is compressed directly with other information, the radiation resolution of more effective information around peak will reduce.

To adapt to the diverse SAR SLC images (stored in Uint16 format) and achieve the delicate feature expression with limited bits, the adaptive normalization method is proposed and realized by means of piecewise mapping, as shown in Fig. 17. The piecewise mapping takes the peak value of the amplitude distribution as the reference to calculate the piecewise thresholds, which can be expressed as [82]

$$\mathrm{NV} = \begin{cases} 1.0, & \mathrm{OA} > (b+r)\cdot \mathrm{Peak} \\ -\dfrac{\mathrm{OA}}{(a+r)\cdot \mathrm{Peak}}, & \mathrm{OA} < (a+r)\cdot \mathrm{Peak} \\ \dfrac{\mathrm{OA}}{(b+r)\cdot \mathrm{Peak}}, & \text{else} \end{cases} \tag{6}$$

where a and b are constants set to 5.5 and 8.5 in this article, Peak denotes the peak value of the amplitude distribution, r is a random factor uniformly distributed between -0.1 and 0.1, and NV and OA denote the normalized value and the original amplitude, respectively. Calculating piecewise thresholds with

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reference to the peak value ensures the relatively uniform distribution of various images, while truncating the long tail directly enhances the image contrast.

In addition, the random perturbance of thresholds is only added during training (as shown in Fig. 17), which enhances the generalization and robustness of detectors for diverse scenarios.

The other important module, namely, the STFT analysis module, aims to extract local spatial correlation information underlying TFD. Compared to intensity images, the incremental information contained in SLC images can assist detectors in distinguishing between objects and backgrounds. Different types of objects exhibit various scattering intensities in response to different frequency components within the processing bandwidth, which can serve as the basis for distinguishing among various objects [31]. Therefore, the SLC imagery is decomposed into components at varying frequencies through the STFT to assist fine-grained OD&R. Fig. 18 displays the decomposition process, which can be described as [83]

$$M(r_0, a_0, f) = \left| F\left\{ h\left(\hat{r} - r_0\right) \cdot m\left(\hat{r}, a_0\right) \right\} \right| \tag{7}$$

where $M(\cdot)$ denotes the magnitude value of the STFT result, $m(\cdot)$ denotes the original SAR SLC image, $F\{\cdot\}$ denotes the operation of Fourier transform, and r_0, a_0, and f denote the coordinates in terms of range, azimuth, and frequency in the TF cube, respectively. h denotes the window applied for STFT, and this article applies the Gaussian windows for the optimal tradeoff between the TD resolution and frequency-domain resolution.

2) Dual Backbone for Feature Extraction and Fusion: As shown in Fig. 16, the dual backbone is proposed to extract features in TD and TFD from SLC SAR images. It consists of the TD feature extraction (TD) backbone and the TFD feature extraction (TFD) backbone.

The TD backbone employs ResNet50 [78] and the feature pyramid network (FPN) [84] to extract essential visual features, such as structure, shape, and texture, from TD images after the abovementioned normalization process. The output of TD backbone is five feature maps with the same channel dimension, namely, 256, and various scales at $\{16 \times 16, 32 \times 32, 64 \times 64, 128 \times 128, 256 \times 256\}$. The large-scale shallow feature maps with high resolution contribute to the OD&R of small-scale subcategories, such as "helicopter" and "Fokker-50." The small-scale deep feature maps with detailed semantic information are beneficial for the refinement and classification of OBB.

The other backbone, namely, the TFD backbone, extracts incremental information from TFD for distinguishing various targets or backgrounds. Its structure is similar to that of the TD backbone, but it replaces ResNet50 with the simpler double convolution layer [78], where layer normalization [85] and ReLU activation are supplemented at each layer. The output of the TFD backbone is also five feature maps with the same scale as that of the TD backbone. However, the channel dimension of them is 128.

After extraction, the above two outputs are fused in the following way, that is, the feature maps from various backbones are concatenated along the channel dimension and further undergo 1×1 convolution. The feature maps after multidomain fusion have the same channel dimension, namely, 256. The fusion feature maps, with both fundamental information from TD and incremental information from TFD, are more beneficial for OD&R, which is the key to performance improvement.

3) Detection Head for OBB Regression and Recognition: The detection head serves to obtain the OD&R results based on the multidomain fusion feature maps. It consists of a region proposal network (RPN), rotated region of interest align (RRoI Align), and OBB refinement and recognition. The RPN proposes regions that may contain objects of interest and provides their anchors. Each anchor is defined by the 5-tuple (x, y, w, h, θ), where (x, y) denotes the center coordinate of the anchor and (w, h, θ) denotes its width, height, and angle. These anchors are then applied for the RRoI Align. The RRoI Align is first proposed in the RoI Transformer algorithm [38]. It maps the rotated region to the multidomain feature maps and performs interpolation to ensure that the extracted features accurately reflect rotated objects. RRoI Align then carries out pooling operations to obtain feature vectors with a fixed size. These feature vectors are finally applied for the OBB refinement and recognition. The OBB refinement module defines each bounding box with the same 6-tuple but provides higher location accuracy with further refinement. The recognition module predicts the posterior probability of each category, and the category with the highest probability is taken as the recognition result.

4) Loss Function: The total loss for model optimization is divided into two components, which are respectively calculated in the region proposal stage and the OBB refinement and recognition stage. It is described as follows:

$$L_{\text{total}} = L_{\text{orpn}} + \lambda \cdot L_{\text{prn}} \tag{8}$$

where λ is a hyperparameter to weight different components. L_{orpn} is calculated in the region proposal stage and consists of the binary classification loss and the localization loss. They respectively guide the RPN to assess whether there are objects within each region and provide more accurate anchors. L_{orpn} is calculated as

$$L_{\text{orpn}} = L_{\text{bcls}}(p, u) + L_{\text{loc}}\left(t, t^*\right) \tag{9}$$

where p denotes the predicted probability of whether the anchor contains objects. Anchors containing objects are assigned $u = 1$, and others are assigned $u = 0$. t and t^* denote the predicted anchor and the ground-truth OBB, respectively.

L_{prn} is calculated in the OBB refinement and recognition stage and consists of the OBB refinement loss and the recognition loss. The OBB refinement loss optimizes the OBB refinement network for more accurate OBB, while the recognition loss aims to improv the recognition ability of the recognition network. L_{prn} is calculated as

$$L_{\text{prn}} = L_{\text{rloc}}\left(t', t^*\right) + L_{\text{cls}}(q, v) \tag{10}$$

where p denotes the predicted probabilities of all categories, v represents the one-hot encoding of the ground-truth category, and t' denotes the refined OBB.

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D. Results and Analyses

The OD&R results of MDNet on SAR images acquired in SL and FSI modes are presented in Tables V and VI, respectively. The proposed algorithm, namely, MDNet, achieves SOTA performance on OD&R in the FAIR-CSAR dataset. Thanks to multidomain information extraction and fusion, MDNet improves mAP by 2.6% and 7.6% for SAR images acquired in SL and FSI modes, respectively. Fig. 19 visualizes its OD&R results in different scenarios, including the airport, port, river, and ocean surface and covering the typical categories in FAIR-CSAR. The confidence score for the visualization is set to 0.3. As shown in Fig. 19, almost all targets are detected without false alarms.

In addition to object recognition, Fig. 20 also presents the detection results of MDNet for comparison with that of the RoI Transformer given in Fig. 15. It can be found that MDNet also performs better in challenging scenarios where most targets missed by other detectors are detected here. It proves that the proposed MDNet has a strong generalization ability to adapt the practical scenarios, including speckle noise, multipath disturbance, azimuth ambiguities, motion defocusing, and so on.

V. Conclusion

A benchmark dataset denoted as FAIR-CSAR is proposed for fine-grained OD&R on complex-valued SAR images. Contributed to its significant advantages, such as enormous quantity (over 340k) instances, fine-grained annotations, diverse scenarios, and data domains, FAIR-CSAR is hoped to be the experimental foundation for potential development and comprehensive validation of SAR physics-aware OD&R algorithms. Moreover, some challenging examples caused by unique SAR imaging mechanisms are also included, which are beneficial for the performance validation in practical scenarios. The relevant illustrations and analyses of FAIR-CSAR, including category design, statistical characteristics, and so on, are also detailed here. A series of SOTA algorithms are applied to build the OD&R benchmark to foster future research. In addition, an MDNet is proposed as a novel framework to mine detailed information underlying SLC SAR images. Thanks to the extraction and supplement of incremental information from extra TFD, the proposed MDNet achieves the SOTA performance, with an mAP of 40.6% in the SL mode and 52.2% in the FSI mode. We hope that the proposed FAIR-CSAR dataset can contribute to the development of SAR physics-aware OD&R in the future.

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